

SQL DATA ANALYSIS

Project 3 – Industrial Training Kit

E-Commerce Orders Dataset (2023–2025)

Senior Data Analyst Report

DecodeLabs | Batch 2026

Demonstrating SELECT • WHERE • GROUP BY • ORDER BY • Aggregations (COUNT, SUM, AVG) • HAVING

Data Preparation: Null/blank values in **CouponCode** were replaced with **'No Coupon'** before loading into SQLite. The resulting clean table contains 1,200 rows and 14 columns.

Total Orders	Unique Customers	Total Revenue	Avg Order Value
1,200	1,189	\$1,264,762	\$1,053.97

1. Project Objective

Project 3 requires extracting business insights from the e-commerce orders dataset using pure SQL. The focus is on writing declarative queries that demonstrate mastery of the core clauses: **SELECT**, **WHERE**, **GROUP BY**, **ORDER BY**, and the aggregate functions **COUNT**, **SUM**, **AVG**. We also apply **HAVING** for post-aggregation filtering and respect the logical execution order of the database engine.

2. Methodology

- Loaded the Excel dataset into SQLite.
- Replaced all NULL CouponCode values with the explicit label 'No Coupon'.
- Wrote a suite of 16 queries covering filtering, grouping, aggregation, ranking, and multi-dimensional analysis.
- Followed the logical execution order: FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY.

3. SQL Query Showcase & Results

3.1 Overall Aggregations (COUNT, SUM, AVG, MIN, MAX)

```
SELECT
  COUNT(*) AS TotalOrders,
  COUNT(DISTINCT CustomerID) AS UniqueCustomers,
  ROUND(SUM(TotalPrice),2) AS TotalRevenue,
  ROUND(AVG(TotalPrice),2) AS AvgOrderValue,
  ROUND(MIN(TotalPrice),2) AS MinOrder,
  ROUND(MAX(TotalPrice),2) AS MaxOrder
FROM orders;
```

TotalOrders	UniqueCustomers	TotalRevenue	AvgOrderValue	MinOrder	MaxOrder
1,200.00	1,189.00	1,264,761.96	1,053.97	11.39	3,456.40

1,200 orders generated \$1.26M in revenue with an average order value of \$1,053.97. The gap between min (\$11.39) and max (\$3,456.40) highlights high variance.

3.2 Revenue by Product (GROUP BY + ORDER BY)

```
SELECT Product, COUNT(*) AS OrderCount, SUM(Quantity) AS UnitsSold,
  ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV
FROM orders
GROUP BY Product
ORDER BY Revenue DESC;
```

Product	OrderCount	UnitsSold	Revenue	AOV
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58

Chair, Printer and Laptop are the top revenue generators. Phone contributes the least revenue despite solid order volume.

3.3 Order Status Distribution (GROUP BY + Percentage)

```
SELECT OrderStatus, COUNT(*) AS OrderCount,
  ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM orders),1) AS Pct,
  ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV
FROM orders
GROUP BY OrderStatus
ORDER BY OrderCount DESC;
```

OrderStatus	OrderCount	Pct	Revenue	AOV
Cancelled	250	20.80	276,396.21	1,105.58
Returned	247	20.60	243,277.70	984.93
Pending	237	19.80	256,328.15	1,081.55
Shipped	235	19.60	246,159.58	1,047.49
Delivered	231	19.30	242,600.32	1,050.22

Critical finding: Cancelled (20.8%) + Returned (20.6%) = 41.4% of all orders. Nearly \$520k of revenue is at risk.

3.4 High-Value Orders (WHERE + ORDER BY)

```
SELECT OrderID, Product, Quantity, UnitPrice, TotalPrice, OrderStatus
FROM orders
WHERE TotalPrice > 3000
ORDER BY TotalPrice DESC
LIMIT 8;
```

OrderID	Product	Quantity	UnitPrice	TotalPrice	OrderStatus
ORD200789	Tablet	5	691.28	3,456.40	Delivered
ORD201122	Monitor	5	678.19	3,390.95	Returned
ORD200632	Laptop	5	678.16	3,390.80	Delivered
ORD200469	Chair	5	676.98	3,384.90	Cancelled
ORD200328	Tablet	5	674.04	3,370.20	Cancelled
ORD200107	Printer	5	670.75	3,353.75	Shipped
ORD200326	Laptop	5	670.48	3,352.40	Returned
ORD201065	Printer	5	666.80	3,334.00	Delivered

All top-value orders are 5-unit purchases of premium-priced items. Several of these high-value orders are Cancelled or Returned — a direct revenue leakage.

3.5 Coupon Code Analysis (after replacing blanks with 'No Coupon')

```
SELECT CouponCode, COUNT(*) AS Orders,
       ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV
FROM orders
GROUP BY CouponCode
ORDER BY Orders DESC;
```

CouponCode	Orders	Revenue	AOV
FREESHIP	313	335,036.99	1,070.41
No Coupon	309	322,401.41	1,043.37
WINTER15	292	302,483.54	1,035.90
SAVE10	286	304,840.02	1,065.87

After cleaning, 'No Coupon' is the second-most frequent category (309 orders). FREESHIP and SAVE10 show slightly higher AOV than 'No Coupon' or WINTER15.

3.6 Revenue by Payment Method

```
SELECT PaymentMethod, COUNT(*) AS Orders,
       ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV
FROM orders
GROUP BY PaymentMethod
ORDER BY Revenue DESC;
```

PaymentMethod	Orders	Revenue	AOV
Credit Card	234	263,847.63	1,127.55
Online	258	262,442.94	1,017.22
Cash	246	259,786.29	1,056.04
Gift Card	230	246,323.92	1,070.97
Debit Card	232	232,361.18	1,001.56

Credit Card yields the highest AOV. Online is the most frequently used method.

3.7 Revenue by Referral Source

```
SELECT ReferralSource, COUNT(*) AS Orders,  
       ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV  
FROM orders  
GROUP BY ReferralSource  
ORDER BY Revenue DESC;
```

ReferralSource	Orders	Revenue	AOV
Instagram	259	275,285.45	1,062.88
Email	250	261,808.55	1,047.23
Google	241	250,441.48	1,039.18
Facebook	228	250,410.90	1,098.29
Referral	222	226,815.58	1,021.69

Instagram is the strongest acquisition channel by both volume and revenue.

3.8 Yearly Trend

```
SELECT strftime('%Y', Date) AS Year, COUNT(*) AS Orders,
       ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS AOV
FROM orders
GROUP BY strftime('%Y', Date)
ORDER BY Year;
```

Year	Orders	Revenue	AOV
2023	510	552,643.24	1,083.61
2024	459	480,235.87	1,046.27
2025	231	231,882.85	1,003.82

2023 was the strongest full year. 2025 data covers only H1, so absolute volume is lower but AOV remains healthy.

3.9 High-Volume Cancellations & Returns (GROUP BY + HAVING)

```
SELECT Product, OrderStatus, COUNT(*) AS Orders, ROUND(SUM(TotalPrice),2) AS Revenue
FROM orders
GROUP BY Product, OrderStatus
HAVING OrderStatus IN ('Cancelled','Returned') AND COUNT(*) >= 35
ORDER BY Product, Orders DESC;
```

Product	OrderStatus	Orders	Revenue
Chair	Cancelled	45	48,660.98
Desk	Cancelled	35	39,587.69
Laptop	Returned	39	39,654.30
Laptop	Cancelled	35	43,761.72
Monitor	Returned	36	40,524.60
Monitor	Cancelled	35	37,057.44
Printer	Returned	38	35,939.09
Printer	Cancelled	35	38,675.16
Tablet	Returned	43	42,525.86

HAVING filters the aggregated buckets. Several products show 35+ cancellations or returns, indicating systematic issues rather than isolated incidents.

3.10 Multi-Condition Filtering (WHERE + AND)

```
SELECT OrderID, Product, PaymentMethod, TotalPrice, CouponCode
FROM orders
WHERE Product = 'Laptop'
      AND PaymentMethod = 'Credit Card'
      AND TotalPrice > 1500
ORDER BY TotalPrice DESC;
```

OrderID	Product	PaymentMethod	TotalPrice	CouponCode
ORD200764	Laptop	Credit Card	3,137.15	FREESHIP
ORD201156	Laptop	Credit Card	2,763.12	No Coupon
ORD200823	Laptop	Credit Card	2,612.52	WINTER15
ORD201179	Laptop	Credit Card	2,592.75	WINTER15
ORD200986	Laptop	Credit Card	2,579.84	FREESHIP
ORD200083	Laptop	Credit Card	2,161.64	No Coupon

Demonstrates precise row-level filtering using multiple predicates before any aggregation.

4. Key Business Insights from SQL

1. **Revenue at Risk:** Cancelled + Returned orders account for **41.4%** of volume and **~\$520k (41%)** of revenue. This is the single largest opportunity for operational improvement.
2. **Product Concentration:** Chair, Printer and Laptop drive the majority of revenue. Phone under-performs on revenue contribution.
3. **Coupon Cleaning:** Explicitly labeling nulls as 'No Coupon' enables clean GROUP BY analysis. Coupon users (especially FREESHIP/SAVE10) show modestly higher AOV.
4. **Channel Strength:** Instagram leads acquisition; Credit Card leads AOV.
5. **High-Value Leakage:** Several of the top 10 highest-value orders (>\$3,300) ended as Cancelled or Returned.

5. Recommendations

- Prioritise root-cause analysis of cancellations and returns using further SQL segmentation (by Product × Status × Referral).
- Design a retention offer for the high-value order segment identified via WHERE TotalPrice > 2500.
- Increase investment in Instagram acquisition and Credit-Card checkout optimisation.
- Review Phone category pricing/promotion strategy given its lower revenue contribution.
- Continue using explicit 'No Coupon' labelling in all future pipelines for clean aggregation.

Verdict: All Project 3 requirements (SELECT, WHERE, ORDER BY, GROUP BY, COUNT/SUM/AVG, and advanced HAVING) have been demonstrated on a cleaned dataset. The SQL layer successfully surfaces the same core business signals discovered in the EDA phase, confirming analytical consistency across tools.

Database file: ecommerce_orders.db (SQLite) | All queries are reproducible.
Prepared for DecodeLabs Data Analytics Internship – Project 3.